Yuxuan Ren

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EDUCATION

Washington University in St. Louis — St. Louis, MO
Ph.D. Candidate, Environmental Engineering — 08/2022 – expected 2027

Peking University — Beijing, China
M.S., Environmental Science — 09/2019 – 06/2022

Shandong University — Jinan, China
B.E., Environmental Engineering — 09/2015 – 06/2019

RESEARCH INTERESTS

I am an atmospheric chemist. My research integrates aerosol mass spectrometry, ground-based measurements, and chemical transport modeling to investigate aerosol composition, sources, and atmospheric processes.

RESEARCH EXPERIENCE

Research Assistant, Washington University in St. Louis — 01/2023 – Present
Advisor: Randall V. Martin

- Characterize aerosol composition using the globally distributed Surface Particulate Matter Network (SPARTAN)
- Evaluate global black carbon emissions with SPARTAN measurements and chemical transport model GEOS-Chem
- Develop an offline aerosol mass spectrometry method for organic aerosol characterization on PTFE filters
- Build a globally consistent organic aerosol dataset for model evaluation

Research Assistant, Peking University — 06/2018 – 06/2022
Advisor: Hefa Cheng

- Characterized heavy metals in particulate matter (PM) across 12 major cities in China
- Assessed the bioaccessibility and health risks of PM-bound heavy metals using simulated physiological fluids

TECHNICAL SKILLS

Atmospheric measurements: HR-ToF-AMS, ICP-MS, GC-MS, HPLC-MS, SMPS, XRF, IC, TOC, UV-Vis

Data analysis, programming, and modeling: Python, R, MATLAB, GEOS-Chem

PUBLICATIONS

h-index: 7; 217 citations (Google Scholar, July 2026)

First-Author Papers

[4] **Ren, Y.**, Martin, R. V., Chen, J.-H., Oxford, C. R., Williams, B. J., Weber, R. J., Xu, L. Development of an Offline Aerosol Mass Spectrometry Method for Organic Aerosol Characterization in a Globally Distributed Surface Particulate Matter Network. *EGUsphere* [preprint], 2026.
https://doi.org/10.5194/egusphere-2026-2663

[3] **Ren, Y.**, Oxford, C. R., Zhang, D., Liu, X., Zhu, H., Dillner, A. M., White, W. H., Chakrabarty, R. K., Hasheminassab, S., Diner, D., Le Roy, E., Kumar, J., Viteri, V., Song, K., Akoshile, C., Amador-Muñoz, O., Asfaw, A., Chang, R. Y.-W., Francis, D., Gahungu, P., Garland, R. M., Grutter, M., Kim, J., Langerman, K., Lee, P.-C., Lestari, P., Mayol-Bracero, O. L., Naidoo, M., Nelli, N., O'Neill, N., Park, S. S., Salam, A., Sarangi, B., Schechner, Y., Schofield, R., Tripathi, S. N., Windwer, E., Wu, M.-T., Zhang, Q., Rudich, Y., Brauer, M., Martin, R. V. Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. *Nature Communications*, 16, 7010, 2025.
https://doi.org/10.1038/s41467-025-62468-5

[2] **Ren, Y.**, Hu, Y., Cheng, H. Sources, Bioaccessibility and Health Risk of Heavy Metal(loid)s in the Particulate Matter of Urban Areas. *Science of the Total Environment*, 947, 174303, 2024.
https://doi.org/10.1016/j.scitotenv.2024.174303

[1] **Ren, Y.**, Luo, Q., Zhuo, S., Hu, Y., Shen, G., Cheng, H., Tao, S. Bioaccessibility and Public Health Risk of Heavy Metal(loid)s in the Airborne Particulate Matter of Four Cities in Northern China. *Chemosphere*, 277, 130312, 2021.
https://doi.org/10.1016/j.chemosphere.2021.130312

Co-authored Papers

[7] Eom, S., Yeo, M. J., Lee, D., Koo, J.-H., Kim, J., **Ren, Y.**, Oxford, C. R., Liu, X., Dillner, A. M., Martin, R. V., Choi, S.-D., Song, C.-K., Park, J., Kim, H., Park, S. S. Contrasting Long-Range Transport Signals of Metropolitan $PM_{2.5}$ in South Korea Detected by SPARTAN Observations. *Atmospheric Pollution Research*, 17, 103088, 2026.
https://doi.org/10.1016/j.apr.2026.103088

[6] Liu, X., Turner, J. R., Mitroo, D., **Ren, Y.**, Oxford, C. R., Liu, W., Martin, R. V. Assessing Attenuation Effects in X-ray Fluorescence Analysis of Light Elements in Mineral Dust. *ACS ES&T Air*, 3, 175–185, 2025.
https://doi.org/10.1021/acsestair.5c00295

[5] Luo, Q., **Ren, Y.**, Sun, Z., Li, Y., Li, B., Yang, S., Zhang, W., Wania, F., Hu, Y., Cheng, H. Characterization of Atmospheric Mercury from Mercury-Added Product Manufacturing Using Passive Air Samplers. *Environmental Pollution*, 337, 122519, 2023.
https://doi.org/10.1016/j.envpol.2023.122519

[4] Wang, X., **Ren, Y.**, Yu, Z., Shen, G., Cheng, H., Tao, S. Effects of Environmental Factors on the Distribution of Microbial Communities Across Soils and Lake Sediments in the Hoh Xil Nature Reserve of the Qinghai-Tibetan Plateau. *Science of the Total Environment*, 838, 156148, 2022.
https://doi.org/10.1016/j.scitotenv.2022.156148

[3] Luo, Z., Xing, R., Huang, W., Xiong, R., Qin, L., **Ren, Y.**, Li, Y., Liu, X., Men, Y., Jiang, K., Tian, Y., Shen, G. Impacts of Household Coal Combustion on Indoor Ultrafine Particles—A Preliminary Case Study and Implications for Exposure Reduction. *International Journal of Environmental Research and Public Health***,** 19, 5161, 2022.
https://doi.org/10.3390/ijerph19095161

[2] Liu, X., Shen, G., Chen, L., Qian, Z., Zhang, N., Chen, Y., Chen, Y., Cao, J., Cheng, H., Du, W., Li, B., Li, G., Li, Y., Liang, X., Liu, M., Lu, H., Luo, Z., **Ren, Y.**, Zhang, Y., Zhu, D., Tao, S. Spatially Resolved Emission Factors to Reduce Uncertainties in Air Pollutant Emission Estimates from the Residential Sector. *Environmental Science & Technology*, 55, 4483–4493, 2021.
https://doi.org/10.1021/acs.est.0c08568

[1] Luo, Q., **Ren, Y.**, Sun, Z., Li, Y., Li, B., Yang, S., Zhang, W., Hu, Y., Cheng, H. Atmospheric Mercury Pollution Caused by Fluorescent Lamp Manufacturing and the Associated Human Health Risk in a Large Industrial and Commercial City. *Environmental Pollution*, 269, 116146, 2021.
https://doi.org/10.1016/j.envpol.2020.116146

SELECTED PRESENTATIONS

[5] Assessing GEOS-Chem Organic Aerosol Simulations with Globally Consistent Ground-Based Measurements. The 12th International GEOS-Chem Meeting, St. Louis, MO, USA, June 2026. (Poster)

[4] Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. American Geophysical Union (AGU) Annual Meeting, New Orleans, LA, USA, December 2025. (Talk)

[3] Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. International Global Atmospheric Chemistry (IGAC) Early Career Seminar, Remote, July 2025. (Invited Talk)

[2] Progress toward Offline Aerosol Mass Spectrometry (AMS) Measurement of Organic Aerosol. Advancing Globally Distributed Air Quality Monitoring Meeting, St. Louis, MO, USA, June 2025. (Talk)

[1] Black Carbon Emissions Generally Underestimated in the Global South as Revealed by Globally Distributed Measurements. The 42nd American Association for Aerosol Research (AAAR) Annual Conference, Albuquerque, NM, USA, October 2024. (Poster)

FIELD EXPERIENCE

[4] Indoor Volatile Organic Compounds (VOCs) Sampling from Rural Residential Solid-Fuel Combustion. Jinzhong, China, November 2020–November 2022.

[3] Second Scientific Expedition to the Qinghai-Tibet Plateau. Hoh Xil World Heritage Site, China, September–December 2019.

[2] Atmospheric Mercury Sampling in a Fluorescent-Lamp Manufacturing Region. Zhongshan, China, July–August 2019.

[1] Chinese National Pollution Source Census: Residential Stove Upgrade Assessment. "2+26" Cities in Northern China, July 2018–November 2019.

TEACHING EXPERIENCE

Washington University in St. Louis, Teaching Assistant — St. Louis, MO
EECE 514 Atmospheric Science and Climate — Spring 2025
EECE 202 Computational Modeling in EECE — Fall 2023
EECE 314 Air Quality Engineering with Lab — Spring 2023

Peking University, Teaching Assistant — Beijing, China
Environmental Geochemistry — Fall 2019, Fall 2020

HONORS, AWARDS, AND FELLOWSHIPS

[5] Annual Research Impact Award, Washington University in St. Louis — 2025

[4] Pivot 314 Fellowship, Washington University in St. Louis — 2024

[3] Award for Scientific Research, Peking University — 2020

[2] Outstanding Graduate, Shandong University — 2019

[1] National Scholarship, Ministry of Education of China — 2016, 2017, 2018

SERVICE

Peer Reviewer	*Atmospheric Chemistry and Physics; npj Climate and Atmospheric Science;* *Environmental Science and Ecotechnology; Journal of Geophysical Research: Atmospheres*
President	Clear Spring Environmental Protection Association, Shandong University (2016–2017)